

UNIVERSITY OF THESSALY
SCHOOL OF ENGINEERING
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Investigation of Machine Vibration Analysis System Design

Diploma Thesis

Vasileios Dimitriou

Supervisor: Karakonstantis Georgios

Month 2022



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ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΙΑΣ

ΠΟΛΥΤΕΧΝΙΚΗ ΣΧΟΛΗ

ΤΜΗΜΑ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ

**Διερεύνηση Σχεδίασης Συστημάτων Ανάλυσης Δονήσεων
Μηχανών**

Διπλωματική Εργασία

Βασίλειος Δημητρίου

Επιβλέπων/πουσα: Καρακωνσταντής Γεώργιος

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Vasileios Dimitriou

Diploma Thesis

Investigation of Machine Vibration Analysis System Design

Vasileios Dimitriou

Abstract

The abstract includes the scientific area and the purpose - subject of the thesis, the methodology, the main steps followed and the main results obtained. The total extent of the abstract will be up to one page. finding and conclusions to be highlighted.

A thesis written in english should include an english and a greek abstract.

Keywords:

term, term, . . . , term

Διπλωματική Εργασία

Διερεύνηση Σχεδίασης Συστημάτων Ανάλυσης Δονήσεων Μηχανών

Βασίλειος Δημητρίου

Abstract

Η περίληψη περιλαμβάνει την επιστημονική περιοχή και το σκοπό - αντικείμενο της εργασίας, τη μεθοδολογία, τα κύρια βήματα που ακολουθήθηκαν και τέλος τα κύρια αποτελέσματα. Η συνολική έκταση της περίληψης θα είναι μέχρι μία σελίδα.

Διπλωματική εργασία γραμμένη στα ελληνικά, θα πρέπει να περιλαμβάνει ελληνική και αγγλική περίληψη.

Λέξεις-κλειδιά:

όρος, όρος, . . . , όρος

Contents

Abstract	x
Abstract	xi
Contents	xiii
List of Figures	xvii
Abbreviations	xix
1 Introduction	1
1.1 Object of the Diploma Thesis	1
1.1.1 The Concept of Predictive Maintenance	2
1.1.2 Contribution	2
1.2 Structure of the Thesis	3
2 Background	5
2.1 Introduction	5
2.1.1 The Scope of Predictive Maintenance	5
2.1.2 Vibration Analysis Instruments	6
2.1.3 Typical Applications of Vibration Analysis	8
2.2 Fundamentals of Bearing Fault Frequencies	9
2.2.1 Unfault and Fault Models	9
2.3 Algorithms and Technologies	12
2.3.1 Algorithm Introduction	12
2.3.2 Signal Processing	12
2.4 Internet of Things	18

2.4.1	Introduction to Internet of Things & Key Technologies	18
2.4.2	Cloud-Oriented Perspective	19
2.4.3	Internet- vs. Things-Oriented Perspectives	19
2.4.4	Semantic-Oriented Approach	20
2.5	Project Implementation: Sensing & Communication	20
2.5.1	The Sensor Specifications	20
2.5.2	The Communication Protocol	22
3	System Engineering and Design	27
3.1	System Requirements	27
3.1.1	Functional Requirements	27
3.1.2	Non-Functional Requirements	28
3.2	Hardware Architecture	28
3.2.1	Component Selection	28
3.3	Software Architecture and Data Flow	28
3.3.1	Signal Processing Pipeline	28
3.3.2	Scalability Considerations	29
3.3.3	Arduino sketch	30
3.3.4	Raspberry Pi coding	30
3.3.5	Vercel coding	30
4	Results	31
4.1	Intro	31
5	Conclusions	33
5.1	Σύνοψη και συμπεράσματα	33
5.2	Μελλοντικές επεκτάσεις	33
	Bibliography	35
	APPENDICES	39
	Appendix	
	Τίτλος Παραρτήματος	41
1	Δυνατότητες του L ^A T _E X	41

1.1	Πίνακες	41
1.2	Διαγράμματα - Γραφικές παραστάσεις	42
1.3	Σχήματα	42
1.4	Αλγόριθμοι	42
1.5	Μαθηματικές εκφράσεις	43
1.6	Θεωρήματα, Πορίσματα, Ορισμοί, κλπ.	43
1.7	Απαριθμήσεις	44
1.8	Είδη πηγών στις αναφορές	44

List of Figures

2.1	The Internet of Things paradigm as a result of the convergence of different visions. Source: [1].	7
2.2	Typical vibrometer set-up.	8
2.3	ADXL 335 accelerometer.	9
2.4	Fault and unfault spectrum graphs [2].	10
2.5	Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.	12
2.6	Data collection and assessment algorithm for bearing failure prediction. Source: [3].	13
2.7	Recursive decomposition using even-odd splitting.	18
2.8	The Internet of Things paradigm as a result of the convergence of different visions. Source: [4].	21
2.9	The UART block diagram. Source: [5].	23
3.1	Hardware architecture with signal flow	29
3.2	End-to-end data flow with protocols	29
.1	Κλιμάκωση χρόνου εκτέλεσης για διάφορες υποδιαιρέσεις του καννάβου .	42
.2	Παράδειγμα σχήματος με εντολές του πακέτου TikZ	43

Abbreviations

βλπ	βλέπε
κ.λπ.	και λοιπά
κ.ο.κ	και ούτω καθεξής
TEI	Τεχνολογικό Εκπαιδευτικό Ίδρυμα
BPF	Band Pass Filter

Chapter 1

Introduction

Motivation and description. Points:

- Importance of maintenance
- Predictive maintenance
- Object of my thesis
- Contribution of my thesis

Εδώ αυτή κάνουμε μια γενική περιγραφή του χώρου εφαρμογής της διπλωματικής. Αναφέρουμε τα χαρακτηριστικά του χώρου και καταλήγουμε στα γενικότερα προβλήματα που αντιμετωπίζει ο χώρος. Η συζήτηση των προβλημάτων θα πρέπει να προϋποθέτει τον αναγνώστη για το τι θα προσπαθήσει να αντιμετωπίσει η διπλωματική, χωρίς ακόμα να αναφερόμαστε συγκεκριμένα στο αντικείμενο της διπλωματικής.

1.1 Object of the Diploma Thesis

Elaboration on a system that predicts the failure of mechanical components based on historical data. This thesis focuses on resolving the issue of unplanned shutdowns due to corrective maintenance of machines that have bearings. Εδώ αναφερόμαστε συγκεκριμένα στο τί θα κάνει η διπλωματική. Αναφέρουμε λεπτομερώς α) τα προβλήματα που θα λύσει (και που ήδη έχουν περιγραφεί γενικά στην προηγούμενη ενότητα), και β) πώς σκοπεύει να τα λύσει. Είναι σημαντικό κάποιος που θα διαβάσει την ενότητα αυτή να καταλάβει σε σημαντικό βαθμό τον σκοπό της διπλωματικής σας και τις τεχνικές δυσκολίες της, χωρίς

να είναι αναγκαίο να δει όλα τα άλλα κεφάλαια. Η ενότητα αυτή θέλει πολύ προσοχή και καλύτερα να τη γράψετε αφού έχετε γράψει όλα τα υπόλοιπα κεφάλαια.

1.1.1 The Concept of Predictive Maintenance

Getting a good grasp on the upcoming required maintenance activities and being aware of what is of utmost importance is vital for the operation of a plant. A grasp on Periodical status checks of different machines is crucial for the maintenance engineers, since this could yield a clear understanding of what priority is at that instance of time. Therefore the prediction of an upcoming machine failure would constitute an invaluable tool.

- Good condition readings
- Periodical readings
- Readings analysis
- online board

1.1.2 Contribution

Εδώ παραθέτουμε αριθμητικά συγκεκριμένες ενέργειες/λύσεις/μεθοδολογίες που παρουσιάζει η διπλωματική και λύνουν τα προβλήματα που υποσχθήκαμε στην προηγούμενη ενότητα ότι θα λύσει η διπλωματική. Συνήθως η υποενότητα αυτή έχει την παρακάτω μορφή:

Η συνεισφορά της διπλωματικής συνοψίζεται ως εξής:

1. Μελετήθηκαν συστήματα κ.λ.π.
2. Υλοποιήθηκαν τρεις αλγόριθμοι υπολογισμού κ.λ.π.
3. Αξιολογήθηκε η επίδοση των αλγορίθμων και βρέθηκε ότι κ.λ.π.
4. Ενσωματώθηκαν οι αλγόριθμοι σε πρότυπο σύστημα κ.λ.π.
5. ...

1.2 Structure of the Thesis

Εδώ περιγράφουμε τα κεφάλαια της διπλωματικής: μία πρόταση για το τί θα έχει κάθε κεφάλαιο. Συνήθως η ενότητα αυτή έχει την παρακάτω μορφή (δεν θα σας πάρει πάνω από μία μεγάλη παράγραφο):

Εργασίες σχετικές με το αντικείμενο της διπλωματικής παρουσιάζονται στο Κεφάλαιο 2. Το Κεφάλαιο ... συζητά θέματα μοντελοποίησης. Στο Κεφάλαιο ... αναπτύσσουμε κ.λ.π.

Για την τελική οργάνωση του κειμένου σας, συμβουλευθείτε τον επιβλέποντα της εργασίας.

Chapter 2

Background

2.1 Introduction

2.1.1 The Scope of Predictive Maintenance

There is a tremendous abundance of applications where the vibration analysis is a prerequisite. Readings and records are of vital importance in the context of the predictive maintenance implementation. Most of the industries nowadays strive to make use of accurate preventive maintenance plans, which requires a well-rounded continuous monitoring system, that keeps data of the whole equipment. The ultimate goal of the preventive maintenance plan is the elimination of the unwanted shut-downs due to a machine failure and the increase of plant reliability.

Most of the machines need bearings because bearings are mechanical components designed to support rotating shafts and minimize friction. The bearings failure may have a wide variety of causes, lubrication shortage, seal failure, high temperatures, misalignment are some of the causes among others. Therefore vibration analysis focuses on predicting all these possible causes of bearings failure. Some of the preventive actions against the bearings failure are described bellow.

Proper lubrication plays a vital role in bearing longevity. Rather than choosing oils based solely on moderate operating temperatures, engineers must consider factors like bearing speed, size, and load. Maintaining an adequate oil film is critical, and minimum viscosities are established based on bearing type. This oil film ensure that no metal-to-metal contact takes place for a predetermined load and speed. If these thresholds aren't met, metal-to-metal

contact occurs, significantly reducing reliability [1].

Alignment is another critical cause of bearings failure. A misalignment of some equipment -typically consisted of a motor and the main equipment- leads to bearings failure. For this purpose vibration and temperature monitoring are significant. For this reason, alignment practice upon machine installation is mandatory. According to Bloch the proper initial shaft alignment is necessary in any case and it is proven that the most advantageous technique to do so is the laser optic one [1].

A detailed alignment procedure is also presented by Bloch. On Figure 2.1 is depicted a typical aligning tool set-up. The laser headers must be installed on machined surfaces, which entails that it will be settled on the motor shaft and the shaft of the rotated mechanism. This laser mechanism has an emitter system, that produces a laser beam, and the laser beams are received by some receivers. Usually a three reference point alignment is what the optic laser machines offer, and this procedure takes place while the motor and the rotated machine are coupled with each other.

The laser mechanism set-up (laser emitters and receivers) are connected to some alignment computer, which is powered by some special software. This yields some reading and some alignment correction recommendations. The alignment criteria depend on the mechanical design of the equipment and the rotating speed as well [1].

However there could be some more causes of machinery failure, specifically a loose machinery base or tightening bolt is another that could lead to a components failure. Hence, the way to alleviate this issue is to align and tighten the machinery back to the frame.

How does vibration diagnosis contribute to operational reliability and cost efficiency in engineering applications? The answer to this question is that the implementation of vibration-analysis-based predictive maintenance helps in the direction of troubleshooting the cause of some issue. In addition, raising awareness on some upcoming failure leads to proper planning, which can prevent greater equipment damage.

2.1.2 Vibration Analysis Instruments

There is a great number of vibration monitoring products. Some of them are listed here-under:

- amplitude velocity and acceleration (range of motion) vibrometers

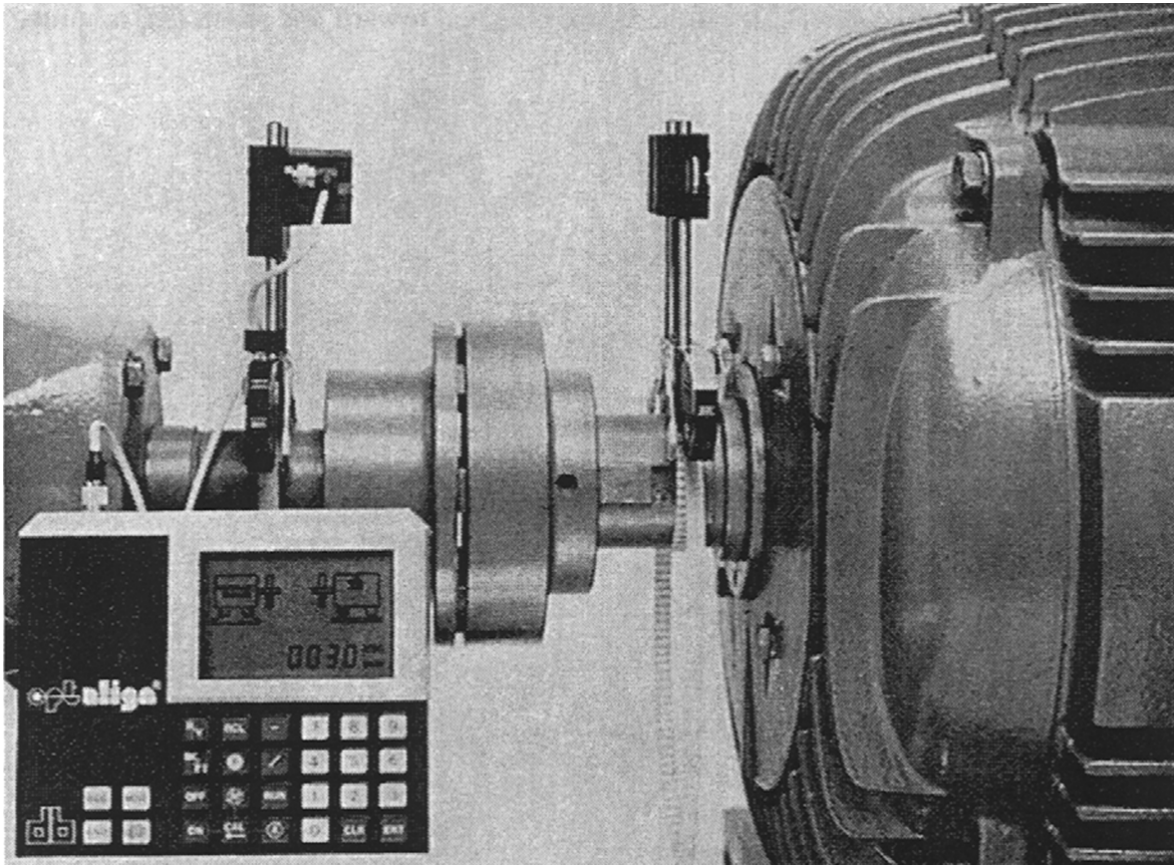


Figure 2.1: The Internet of Things paradigm as a result of the convergence of different visions.
Source: [1].

- frequency (rate of oscillation repetition) vibrometers
- accelerometers

The accelerometers and vibrometers, both can be remote - cordless sensor units or wired sensor units connected to some computer. The latter device is a smart device powered by some software so as to store and analyze data. In many cases these devices may also offer some online-cloud options, storing data on some database yielding accessibility for multiple users.

The device depicted in the image exemplifies a standard configuration of a vibration meter and an accelerometer (see Figure 2.2).

The accelerometer measures vibration events in mg (or in some cases in mm/s^2 and interprets these events into frequency domain data. While the vibrometer stores reading of vibration velocity (mm/s) of vibration acceleration (mm/s^2).

These devices have the following common traits, they require the machinery size and



Figure 2.2: Typical vibrometer set-up.

rotation as an input, so as to proceed with the results extraction. In addition to this, these can execute this calculation on all three directions simultaneously. And to top it all off, they are expensive.

There is some more accessible alternatives. Specifically, there is some small sized low-cost acceleration sensor on breakouts that can be used on related projects (see Figure 2.3).

2.1.3 Typical Applications of Vibration Analysis

Some typical applications that use vibration analysis implementations are turbine motors and shafts, motors connected pumps and motors connected to roller equipment among others.

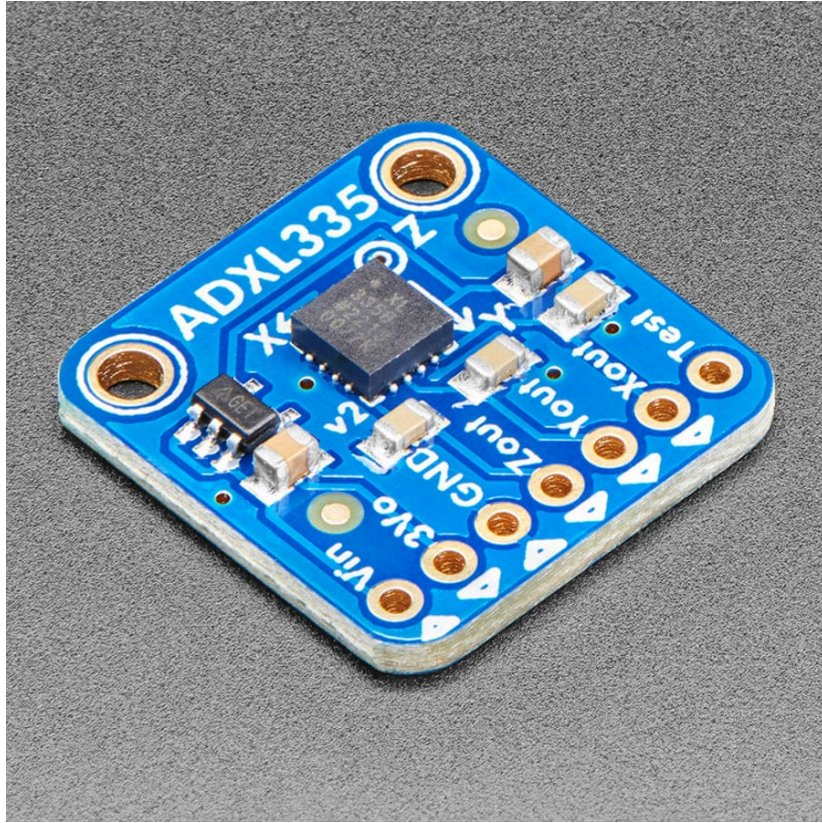


Figure 2.3: ADXL 335 accelerometer.

2.2 Fundamentals of Bearing Fault Frequencies

The most significant challenge that the authors had to deal with is that of the faulty cases detection. For this reason, fault and unfault models have been developed, and series of comparisons between the actual model generated by apparatus measurements and these two situational models are being executed. This development constitutes a real challenge since this is the main criterion that identifies the state of the motor.

2.2.1 Unfault and Fault Models

As previously mentioned, the authors make use of a model-based approach, which entails that the main concept is comparison-oriented. Specifically, there is the unfault model, which contains a high-amplitude peak at the shaft rotating frequency F_ω . In addition to that, it can be noticed in Figure 2.4 that some other patterns distinct themselves from the previously-mentioned unfault model. Regarding the unbalance model, it is obvious that the amplitude at the first tone is much greater. In the case of the misalignment model, it can be noticed that the tone at $2F_\omega$ has a much greater amplitude while the amplitude at the tone of the

shaft speed F_ω remains at a high level. In the case of looseness, there are some more tones prevailing on the spectrum, among the tones related to the shaft speed. To top it all off, the most important vibration spectrum is that of the bearing failure (Refer to Figure 2.5). Every bearing is characterized by four critical frequencies, given by its manufacturer, and these frequencies are associated with the four bearing components (inner and outer ring, balls, and cage):

- BPFO (Ball Pass Frequency Outer) - outer race failure
- BPFI (Ball Pass Frequency Inner) - inner race failure
- BSF (Ball Spin Frequency) - rolling element failure
- FTF (Fundamental Train Frequency) - cage failure

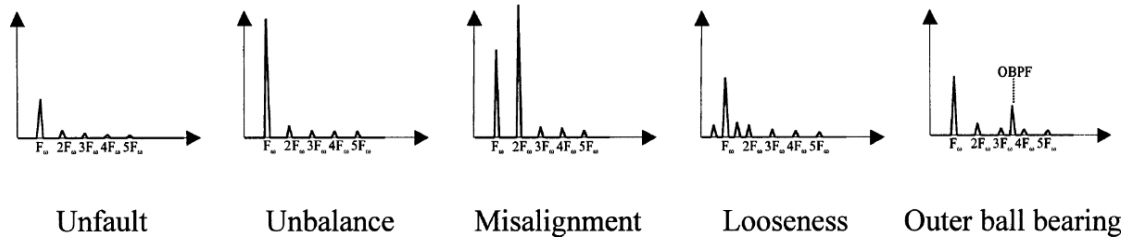


Figure 2.4: Fault and unfault spectrum graphs [2].

The procedure to calculate critical frequencies involves two steps: first computing the bearing frequency factors, then deriving the vibration frequencies. The following analysis covers the case where the inner race rotates while the outer race remains stationary.

$$F_{OR} = \frac{Z}{2} \left(1 - \frac{d}{D} \cos \alpha \right) \quad (\text{Outer Race Fault}) \quad (2.1)$$

$$F_{IR} = \frac{Z}{2} \left(1 + \frac{d}{D} \cos \alpha \right) \quad (\text{Inner Race Fault}) \quad (2.2)$$

$$F_{FTF} = \frac{1}{2} \left(1 - \frac{d}{D} \cos \alpha \right) \quad (\text{Cage Frequency, Inner Race Rotating}) \quad (2.3)$$

$$F_{BS} = \frac{D}{d} \left[1 - \left(\frac{d}{D} \right)^2 \cos^2 \alpha \right] \quad (\text{Ball Spin Frequency}) \quad (2.4)$$

Then, the aforementioned factors when multiplied with shaft speed (f) give specific critical bearing vibration frequencies:

$$\text{BPFO} = f \times F_{\text{OR}} \quad (2.5)$$

$$\text{BPFI} = f \times F_{\text{IR}} \quad (2.6)$$

$$\text{BSF} = f \times F_{\text{BS}} \quad (2.7)$$

$$\text{FTF} = f \times F_{\text{FTF}} \quad (2.8)$$

where:

f	: Shaft rotational speed (Hz)
BPFI	: Ball pass frequency, inner race
BPFO	: Ball pass frequency, outer race
FTF	: Fundamental train frequency
BSF	: Ball spin frequency
Z	: Number of rolling elements
D	: Pitch circle diameter of the bearing (mm)
d	: Rolling element (ball) diameter (mm)
α	: Contact angle ($^{\circ}$)

It is also worth noting that the bearing frequency factors can be obtained through:

- Manufacturer-provided values
- Direct calculation using the above equations (2.1)–(2.4)
- Reverse calculation from measured vibration frequencies (2.5)–(2.8)

To summarize the preceding analysis, Figure 2.5 provides a clear visualization of the fault diagnosis procedure. When analysis reveals elevated frequency tones, a systematic approach must be followed to identify the source of these dominant frequencies. This process essentially involves mapping characteristic vibration patterns to specific machine faults on a frequency spectrum plot.

It must be that a fault-emulation method can be used in such cases to simulate the faulty models. This entails that additional tones can be stressed, and crucial spectra modifications may take place for research and comparison purposes.

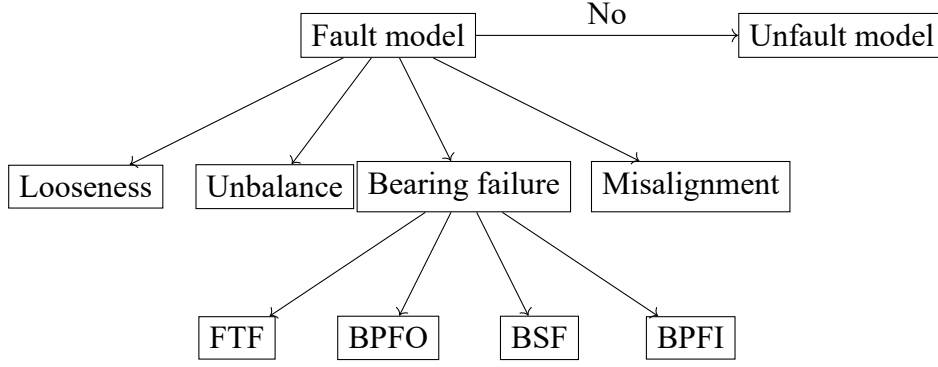


Figure 2.5: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

2.3 Algorithms and Technologies

2.3.1 Algorithm Introduction

This section aims at yielding a brief introduction into the concept of bearing failure predictive maintenance plan. The set-up utilizes an accelerometer which collects data that are going to be used for model assessment, by the means of Fast-Fourier Analysis.

In most of the actual applications a machine learning implementation is a common practice. A figure that depicts this modeling has been presented by Dharmarathne et al [3].

2.3.2 Signal Processing

Fourier Analysis

In this section a deeper elaboration on Fourier Analysis and Transform is going to take place.

Serafeimidou explains in detail the theoretical background of the Fourier Analysis [6]. For an arbitrary finite range $[-L, L]$ there is a family of functions, which meet the Dirichlet condition, and these functions can be expressed as a trigonometric sequence:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\pi\omega x) + b_n \sin(n\pi\omega x)), \quad (2.9)$$

where $\omega = \frac{\pi}{L}$.

In addition its constants can be expressed as follows:

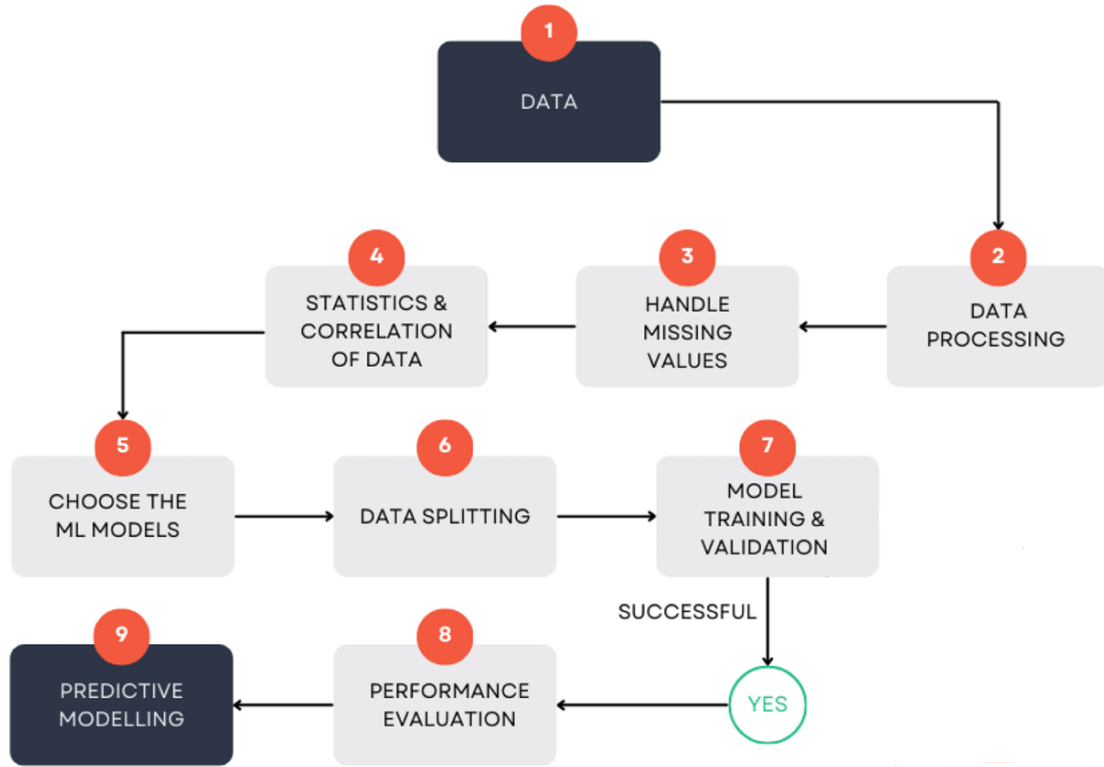


Figure 2.6: Data collection and assessment algorithm for bearing failure prediction. Source: [3].

$$a_n = \frac{1}{L} \int_{-L}^L f(u) \cos(n\omega u) du \quad (2.10)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(u) \sin(n\omega u) du \quad (2.11)$$

The equation (2.9) needs to be expressed as a valid form within the whole $x \in \mathbb{R}$. For this reason the trigonometric integral (which is called Fourier integral) can be expressed as follows:

$$F(x) = \int_0^\infty [A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)] d\omega \quad (2.12)$$

where:

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(u) \cos(\omega u) du \quad (2.13)$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^\infty f(u) \sin(\omega u) du \quad (2.14)$$

Assuming that there exist real functions $f(x)$ defined in $\mathbb{R}$ meeting the following points:

- $f(x)$ and its derivative $f'(x)$ are continuous in the range $[-L, L] \subset \mathbb{R}$.
- $f(x)$ is integrable in $\mathbb{R}$:

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty.$$

Hence, given that the aforementioned conditions are met, $A(\omega)$ (see Eq. (2.13)) and $B(\omega)$ (see Eq. (2.14)) tend to attain a real value for ω , which implies that $A(\omega)$ and $B(\omega)$ are defined for every $\omega \in \mathbb{R}$. In addition to this, the integral of the equation (2.12) tends to attain a real value for x , which ensues that $F(x)$ is defined for every $x \in \mathbb{R}$.

$F(x)$ is identical to $f(x)$ whenever $f(x)$ is a continuous function. There are some special cases when $f(x)$ is not a continuous function, and to deal with these cases, we define:

$$F(x) = \frac{1}{2} \left[\lim_{x \rightarrow x^+} f(x) + \lim_{x \rightarrow x^-} f(x) \right] \quad (2.15)$$

This ensures that $F(x)$ is properly handled even in cases of discontinuity.

All things considered the Fourier Transform $F(\omega) = \mathcal{F}\{F(x)\}$ can be mathematically expressed as follows:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (2.16)$$

where the frequency can derive from the following form: $\omega = \frac{2\pi jk}{N}$.

In practice, the Discrete Fourier Transform (DFT) boasts an extremely wide variety of applications, facilitating the Fourier Transform when applied to sampled signals.

Hence, the mathematical formulation for determining the amplitude of each frequency, given a waveform $x_0, x_1, \dots, x_{n-1}$ consisting of real values, is structured as follows:

$$X(j) = \sum_{k=0}^{n-1} x(k) e^{-\frac{i2\pi jk}{n}}, \quad \text{for } j = 0, 1, \dots, n-1. \quad (2.17)$$

Fast Fourier Transform (FFT)

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform (DFT), expressed by the equation (2.17) and its inverse. One of the most prominent FFT algorithms, the Cooley-Tukey algorithm [7], employs a divide-and-conquer approach to reduce the computational complexity from $\mathcal{O}(n^2)$ (for the naive DFT) to $\mathcal{O}(2n \log n)$, enabling significant speedups for large amount of data.

This reduces complexity from $\mathcal{O}(n^2)$ (naive DFT entails direct implementation of the equation (2.17)) to less than $\mathcal{O}(2n \log n)$ (a Fast Fourier Transform implementation), enabling major time savings for complex and large-scale transforms. It is also worth articulating that the complexity can be interpreted as the number of operations executed [7], [8].

Both algorithms are implemented in major numerical packages including MATLAB and NumPy's routines [9]. What is highlighted by Singleton is that the Cooley-Tukey approach is considerably faster and decently performing especially when coupled with some additional practices so as to execute the Fast Fourier Transform [8].

The computation time according to the aforementioned article decreases drastically by making use of the Cooley-Tukey algorithm. In addition, this very algorithm performs and adapts better to different applications compared to Good's, Danielson's and Goertzel's approaches.

Cooley and Tukey proved that the sample length can be composite size where it is preferable to make use of sample points sized as power of 2, especially when it comes to personal computers with binary arithmetic system. In addition to this the authors have proven that the power of 3 is formally the most efficient approach, however in the context of the fast Fourier Analysis the so-called Radix-2 implementation is the prominent one [7].

$$N = 2^m \quad (2.18)$$

It is hereby worth it elaboration on the Fast Fourier Transform algorithm [7]. The indices of the equation (2.17) can be expressed as follows:

$$\begin{aligned} j &= j_{m-1} \cdot 2^{m-1} + j_{m-2} \cdot 2^{m-2} + \dots + j_1 \cdot 2 + j_0, \\ k &= k_{m-1} \cdot 2^{m-1} + \dots + k_1 \cdot 2 + k_0, \end{aligned} \quad (2.19)$$

where j_v and k_v are the contents of the respective bit positions within the binary representation, which ensues that they are equal to 0 or 1.

The aforementioned set-up (see eq. (2.17)) leads to the following equation (2.21), encompassing twiddle factors expressed as:

$$W = e^{-2\pi i/N} \quad (2.20)$$

And the Fast Fourier Transform:

$$X(j_{m-1}, \dots, j_0) = \sum_{k_0} \sum_{k_1} \dots \sum_{k_{m-1}} x(k_{m-1}, \dots, k_0) \cdot W^{j \cdot k_{m-1} \cdot 2^{m-1} + \dots + j \cdot k_0} \quad (2.21)$$

where as per the first step:

$$W^{j \cdot k_{m-1} \cdot 2^{m-1}} = W^{j_0 \cdot k_{m-1} \cdot 2^{m-1}} \quad (2.22)$$

Furthermore, in the context of the first calculation, the first array can be expressed as follows:

$$x_1(j_0, k_{m-2}, \dots, k_0) = \sum_{\mathbf{k}_{m-1}} x(\mathbf{k}_{m-1}, \dots, \mathbf{k}_0) \cdot W^{j_0 \cdot \mathbf{k}_{m-1} \cdot 2^{m-1}} \quad (2.23)$$

And hereunder can be contemplated that proceeding to the next innermost-successive sum:

$$W^{j \cdot k_{m-1} \cdot 2^{m-1}} = W^{(j_{l-1} \cdot 2^{l-1} + \dots + j_0) \cdot k_{m-l} \cdot 2^{m-l}} \quad (2.24)$$

followed by the calculation of the arrays:

$$x_l(j_0, \dots, j_{l-1}, k_{m-l-1}, \dots, k_0) = \sum_{\mathbf{k}_{m-1}} x_{l-1}(j_0, \dots, j_{l-2}, \dots, k_{m-l}, \dots, k_0) \cdot W^{(j_{l-1} \cdot 2^{l-1} + \dots + j_0) \cdot k_{m-l} \cdot 2^{m-l}} \quad (2.25)$$

for $l = 1, 2, \dots, m$

And this could constitute a recursive procedure obtaining the successive arrays. And last but not least, this finally could lead to the calculation of X spectrum. It can be contemplated that from the Equation (2.25), there is at all times two consecutive values, the even and the odd value that can be stored at a specific location corresponding to an index of:

$$(j_0 2^{m-1} + \dots + j_{l-1} 2^{m-l} + k_{m-l-1} 2^{m-l-1} + \dots + k_0) \quad (2.26)$$

It is also worth it recalling that m is defined by $N = 2^m$. Since bit positions $2^{(m-l)}$, meaning the j and k coefficients, can only get the values 0 and 1, which ensues that there is multiple-storage locations that can be calculated simultaneously.

The array calculated gives the desired Fourier sums:

$$X(j_{m-1}, \dots, j_0) = A_m((j_0, \dots, j_{m-1})) \quad (2.27)$$

At this point there is a need to simplify the Fast Fourier Transform algorithm. It is mandatory to recall the (2.21) so as to depict the algorithm.

The calculations workload can be reduced, by decomposing the original input sequence $X[k]$, of length N , into two subsequences:

- A sequence consisting of elements at even-numbered indices: $X_{\text{even}}[k]$
- A sequence consisting of elements at odd-numbered indices: $X_{\text{odd}}[k]$

By processing these smaller sequences independently, the algorithm substantially reduces the computational workload. The FFT is applied recursively to each half, and this decomposition continues until each subproblem has length 1, at which point the computation becomes trivial (the amplitude value corresponding to a specific frequency). Furthermore, the results from the smaller FFTs are recombined using twiddle factors (see Equation (2.20)).

An significant property of this algorithm is that that, in the case of $j > \frac{n}{2}$, the values are repeated due to the sinusoidal property of $\exp$, which is known as the symmetry identity. This entails that for the values $j = 0, 1, \dots, n/2 - 1$:

$$X_{\text{even}}(j + n/2) = X_{\text{even}}(j) \quad \text{and} \quad X_{\text{odd}}(j + n/2) = X_{\text{odd}}(j) \quad (2.28)$$

The evens and odds decomposition:

$$X_{\text{even}}(j) = \sum_{k=0}^{n/2-1} x(2k) e^{-\frac{i2\pi jk}{n/2}}, \quad \text{for } j = 0, 1, \dots, n/2 - 1. \quad (2.29)$$

$$X_{\text{odd}}(j) = \sum_{k=0}^{n/2-1} x(2k+1) e^{-\frac{i2\pi jk}{n/2}}, \quad \text{for } j = 0, 1, \dots, n/2 - 1. \quad (2.30)$$

So from the Equation (2.21):

$$X(j) = X_{\text{even}}(j) + e^{-\frac{i2\pi j}{n}} X_{\text{odd}}(j) \quad (2.31)$$

And this procedure is scalable since each and every X_{even} and X_{odd} can be depleted into a subsequent set of X_{eve} and X_{odd} to the point that the length 1 is reached.

Last but not least, the symmetry identity as described by the Equation (2.28), leads to the following set of equations (see also (2.31)):

$$X(j + n/2) = X_{\text{even}}(j) + e^{-\frac{i2\pi(j+n/2)}{n}} X_{\text{odd}}(j) \quad (2.32)$$

For the values $j = 0, 1, \dots, n/2 - 1$.

The objective of this algorithm is to get to the point that the length of the sums is 1, meaning that the sums of the Equations (2.29) and (2.30) are applicable for $k=0$. The only remaining exponential terms $e^{-\frac{i2\pi j}{n}}$ in the Equation (2.31) - as previously described by Equation (2.25).

The aforementioned procedure can be further repeated, by breaking the branches problem into sub-branches. The radix-2 DIT algorithm to be formulated as follows on Figure 2.7, depicting the concept of narrowing down the problem.

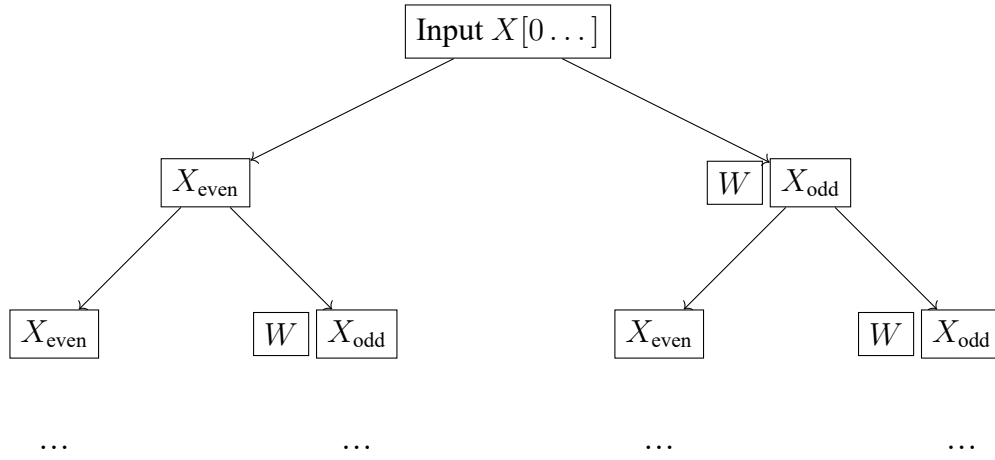


Figure 2.7: Recursive decomposition using even-odd splitting.

All in all, this section yields an overview of utmost importance highlighting the most invaluable Fourier Transform algorithms expressed by equation (2.17) and equation (2.21). As discussed in this section, the Cooley-Tukey Fast Fourier Transform (FFT) algorithm significantly accelerates the computation of the Discrete Fourier Transform (DFT) by recursively breaking the problem into smaller, more manageable subproblems. This divide-and-conquer approach is fundamental to the algorithm's performance advantage.

2.4 Internet of Things

2.4.1 Introduction to Internet of Things & Key Technologies

The ultimate objective of this project is to create a set-up that is well-rounded and has a real-time monitoring and assessment competency. Two of the most prominent technologies that are being incorporated by Internet of Things (IoT) is the cloud computing and the

wireless communication. It is also worth mentioning that, Internet of things encompasses interconnection of physical devices and applications making the most of sensors, networks and data analysis.

2.4.2 Cloud-Oriented Perspective

An extremely accurate definition of the *Internet of Things* is given by Gubbi et al, articulating that:

“Interconnection of sensing and actuating devices providing the ability to share information across platforms through a unified framework, developing a common operating picture for enabling innovative applications. This is achieved by seamless ubiquitous sensing, data analytics and information representation with Cloud computing as the unifying framework.”

This approach is mostly “Cloud”-oriented. The authors focus mostly on the concept of ubiquitous information and evolving communication networks. The means to give ground to this very concept is Smart Connectivity. It is fueled by the prevalence of devices enabled by open wireless technology such as Bluetooth, radio frequency identification (RFID), Wi-Fi, and telephonic data services as well as embedded sensor.

Nevertheless, Hardware-sensor breakouts are critically important, as they serve as the primary means of data collection in IoT systems. Middleware and computing tools for data analytics are equally invaluable, as they enable the processing of raw sensor data into actionable insights. Finally, visualization tools and dashboards play a key role in end-user interaction with the IoT environment, while also providing stakeholders with an intuitive and accessible representation of the data [10].

2.4.3 Internet- vs. Things-Oriented Perspectives

As regards the approach of Atzori et al, there is given a different definition by considering two different approaches, the “Internet”-oriented approach and the “Things”-oriented approach. The “Things”-oriented approach targets mainly at Big Data applications, since according to Atzori et al the time, the place and multiple recipients are the most important factors of this approach. In essence the “Things” that contribute to this perspective are the Radio-

Frequency IDentification (RFID), Near Field Communications (NFC) and Wireless Sensor and Actuator Networks (WSAN) among others.

The "Internet"-oriented vision given by Atzori et al, it is highlighted that the IP (Internet Protocol) constitutes the core technology of Internet of Things (IoT). The Internet Protocol (IP) is lightweight, scalable, and capable of connecting smart objects worldwide—even on low-power devices. The authors also introduce Internet Ø, a similar approach that further simplifies IP to enable "IP over anything," ensuring seamless IoT integration. The ultimate goal of both strategies is to make all objects addressable, reachable, and interoperable through IP, moving from a basic "Internet of Devices" to a fully connected "Internet of Things" [4].

2.4.4 Semantic-Oriented Approach

In addition to the aforementioned perspectives there is an emerging need for representation and storage of the invaluable information, which leads to the "Semantic"-oriented approach (see 2.8). This final approach creates a well-rounded Internet of Things (IoT) concept, yielding more details on it.

The whole concept of the vast majority of different projects requires a semantic approach so as to set some thresholds and yield some feedback or take some action over something. In addition to this the semantic vision according to Atzori et al expresses the need for a representation of the collected data [4].

All in all Atzori defines Internet of Things (IoT) as follows:

"Internet of Things constitutes a world-wide network of interconnected objects uniquely addressable, based on standard communication protocols."

2.5 Project Implementation: Sensing & Communication

2.5.1 The Sensor Specifications

As it has been highlighted by the Internet of Things definitions, the sensors are among the most important devices of this technology. The sensing unit in this project utilizes an ADXL335 accelerometer to collect physical acceleration data, which connects to the micro-controller through an analog input interface.

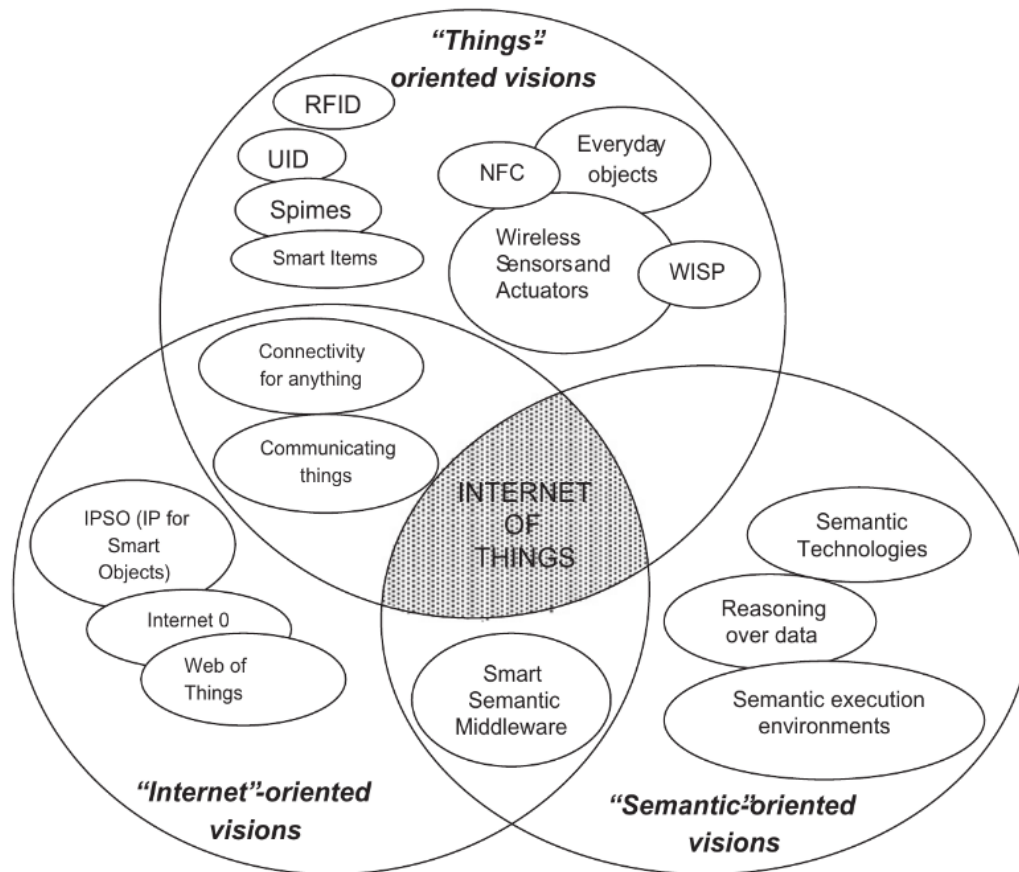


Figure 2.8: The Internet of Things paradigm as a result of the convergence of different visions.
Source: [4].

The accelerometer ADXL335 is a single component sensing X, Y and Z, having a wide measuring range in all three axes. Typical applications of this sensor is structure deflection projects. This sensor is the ideal choice when it comes to vibration analysis applications, since it retains a measuring range of $\pm 3g$. This range adequate with respect to vibration analysis because in most of the rotating equipment applications an acceleration greater than 1,5g indicates a defective component. This entails that the $\pm 3g$ is sufficient in the context of this project. In addition to this, it offers higher resolution, up to 300mV/g for vibration detection, which means it has relatively high sensitivity to small vibrations. As regards its frequency bandwidth, it ranges 0.5-1600Hz. As regards its noise density, it equals to $150 \mu g/\sqrt{Hz}$, which is ideal for low-amplitude vibration waveforms[11].

The Adafruit ADXL335 is a high performance sensor breakout, utilizing capacitors so as to decouple the noise of the power supply and filters om analog outputs for antialiasing and noise reduction.

As for the sensor's theory of operation [11]:

ADXL335 is a polysilicon surface-micromachined structure built on top of a silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration forces. Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and plates attached to the moving mass. The fixed plates are driven by 180° out-of-phase square waves. Acceleration deflects the moving mass and unbalances the differential capacitor resulting in a sensor output whose amplitude is proportional to acceleration. Phase-sensitive demodulation techniques are then used to determine the magnitude and direction of the acceleration.

This combination of mechanical design and signal processing provides robust acceleration measurement without requiring temperature compensation circuits -since this breakout retains a large tolerance into operation temperatures- or complex calibration procedures [11].

2.5.2 The Communication Protocol

The system architecture of the ADXL335 breakout supports analog communication protocol option, however there is a number of different communication protocols frequently utilized by the Adafruit breakouts:

- **Analog Input:** Currently implemented for direct sensor interfacing (ADXL335)
- **Digital Input:** Available for threshold-based detection applications or PWM
- **SPI (Serial Peripheral Interface):** Suitable for high-speed (up to 10 Mbps) communication with multiple peripherals
- **I²C (Inter-Integrated Circuit):** Enables multi-device communication using only two wires (SDA/SCL)
- **UART (Universal Asynchronous Receiver-Transmitter):** Implemented at a predetermined baud rate for Arduino-to-Raspberry Pi communication

The Universal Asynchronous Receiver-Transmitter Protocol

The Arduino board is connected to the Raspberry Pi microcomputer via its USB port. The USB port of the Arduino is in essence a UART device, that has two options, not working simultaneously. The first one out of the two is the Rx - Tx pins option and the second one is the USB-cable option. In the case of the Arduino-Raspberry Pi interconnection, the desired set-up requires a USB-cable.

The Universal Asynchronous Receiver-Transmitter (UART) is a peripheral device for asynchronous serial communication. The baud rate and the data format is configurable, based on the project needs. After hands-on try and error, it has been found out that a baud rate of 115200 performs adequately offering a higher speed on data transferring, which is vital for the sake of this very project. The data transferring speed is of vital importance, since the higher the amount of data sent to the database is, the larger the frequency domain data range can be.

The raw UART (transmitted via USB) data is in a binary format, and upon receipt by the Raspberry Pi the data batch needs to be converted into a string format. For this reason it is mandatory to use UTF-8 (8-bit Unicode Transformation Format) to handle ASCII characters at this point.

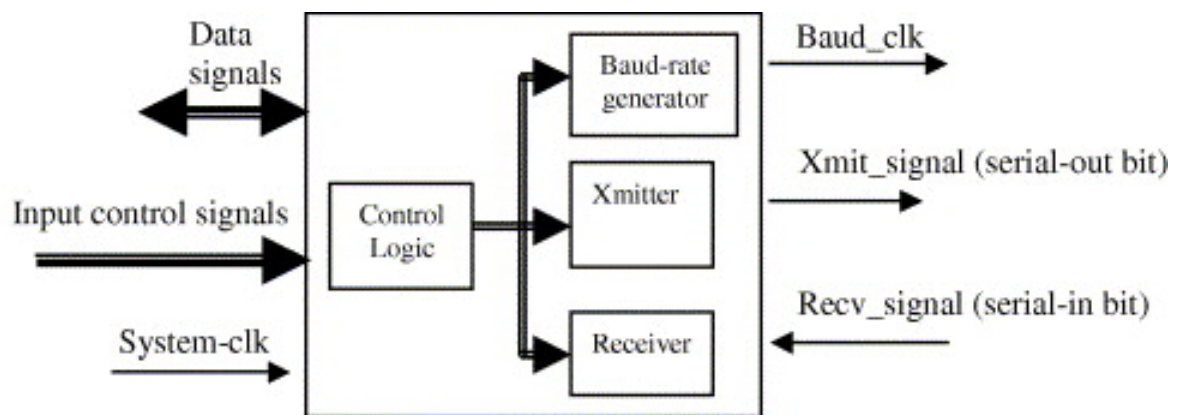


Figure 2.9: The UART block diagram. Source: [5].

A standard UART includes the following key modules (see Figure 2.9):

- **Transmitter** (parallel-to-serial conversion)
- **Receiver** (serial-to-parallel conversion)
- **Baud-rate generator**

- **Control logic**

The baud-rate generator derives a *baud clock* from the system clock. This clock typically operates at 16 times the desired baud rate:

$$\text{baud_clock} = 16 \times \text{baud_rate}$$

This oversampling approach ensures precise timing for bit detection and improves communication reliability.

UART transmits data in structured frames composed of the following elements:

- **Start bit:** Always logic 0
- **Data bits:** Typically 6 to 8 bits (configurable)
- **Optional parity bit:** For basic error detection
- **Stop bits:** Logic 1, usually 1 or 2 bits

The receiver “synchronizes” its internal clock to that of the transmitter’s at the beginning of every data packet received. This allows the receiver to sample the data bit at the bit-cell center. A data packet is composed of 1 start-bit, which is always a logic 0, followed by a programmable number of data bits (typically between 6 and 8). The receiver detects the start bit by sensing a falling edge (logic 1 to 0 transition) and then samples each subsequent bit at the center of its bit period, using the internally generated baud clock. The stop bit must always be logic 1.

UART’s simplicity and robustness make it a widely used solution for embedded systems and serial communication with peripheral devices [5].

Analog Protocol

The Arduino ATmega controllers used for the Arduino contain an onboard 6 channel analog-to-digital (A/D) converter. The converter has 10 bit resolution, returning integers from 0 to 1023. While the main function of the analog pins for most Arduino users is to read analog sensors, the analog pins also have all the functionality of general purpose input/output (GPIO) pins (the same as digital pins 0 - 13) [12].

In this context, it is suggested that the Arduino `analogRead` function work along with some source code that executes calibration function before starts gathering the actual data.

This would increase the reliability of the readings with a small toll on the amount of data spent on measuring. The reason why mapping is required is that the Arduino can only return integers from 0 to 1023. Arduino analog pins compare the input voltage on each pin to a reference voltage, which in turn is the operating voltage 5V. So this practice interprets the measurement input from the sensor as a value within the range [0-1023].

$$\text{analogRead() = } \left(\frac{V_{\text{reading}}}{V_{\text{reference}}} \right) \times 1023 \quad (2.33)$$

Another practice that can also increase the reliability of the readings is the usage of some statistical means that would decrease the chances of reading false data. Some exquisite practices are the average value, the median value and the average value after getting rid of the minimum and maximum values.

The aforementioned practices are not mandatory for this context. In addition to this they reduce severely the amount of data within the reading, which leads to smaller amount of data on frequency domain, after carrying out the Fast Fourier Transform. For this reason, raw data that can go through some mapping-calibration is the ideal practice.

Chapter 3

System Engineering and Design

3.1 System Requirements

The system is designed for low-cost, real-time vibration monitoring of rotating machinery. This set-up encompasses components that are of major importance and its aim is to collect data within a specific - prerequired range, store data safely at a database and finally process data.

The following sections include more detailed overview of the system requirements. Hereunder, more details will be elaborated.

3.1.1 Functional Requirements

The most commonly rotating speed of machinery goes up to 1500 [rpm] corresponds to 25 [Hz]. This relationship is defined by the following equation:

$$f \text{ [Hz]} = \frac{r \text{ [RPM]}}{60} \quad (3.1)$$

The aforementioned relationship entails that the rotational speed of a machine can be interpreted in frequency, which in turn is of great help when related to Fourier Transform applications.

All in all, the functional requirements are highlighted hereunder:

- **Frequency Range:** 0–300 Hz (covers common machine faults like imbalance, misalignment).
- **Real-Time Processing:** On-the-fly FFT computation for immediate fault detection.

- **Scalability:** Modular design for multi-sensor deployments.

3.1.2 Non-Functional Requirements

- **Cost:** Minimize BOM cost (justifies Arduino + RPi4 over industrial PLCs).
- **Power:** Optimize for continuous operation (e.g., no active cooling).
- **Accuracy:** ADXL335's $\pm 3g$ range suffices for industrial vibrations (cite datasheet).

3.2 Hardware Architecture

3.2.1 Component Selection

- **ADXL335 Accelerometer:**
 - Analog output simplifies Arduino ADC interfacing.
 - 300Hz bandwidth meets requirements (vs. digital sensors like MPU6050 needing I2C).
- **Arduino Uno:**
 - Handles analog sampling at 600Hz (Nyquist-compliant for 300Hz signals).
 - Low-latency preprocessing (e.g., DC removal, windowing).
- **Raspberry Pi 4:**
 - Runs Python-based FFT (e.g., NumPy) and SQLite for local storage.
 - WiFi/Bluetooth enables cloud integration (e.g., AWS IoT, InfluxDB).

3.3 Software Architecture and Data Flow

3.3.1 Signal Processing Pipeline

The Arduino samples data at 600Hz, which is extremely high and constitutes a sufficient sample rate for this project. Additionally, another consideration is receiving data from the

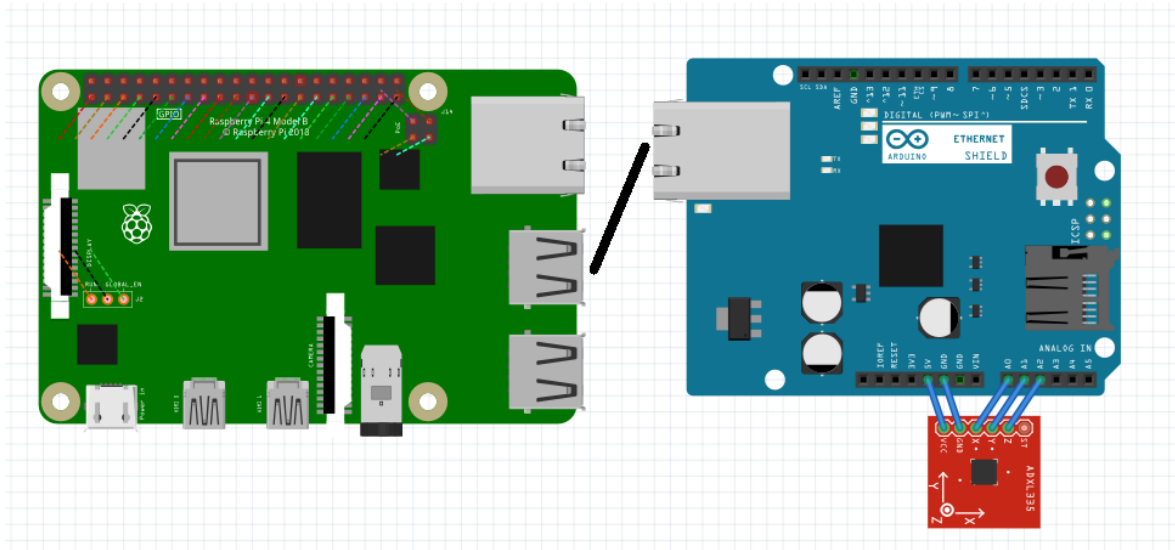


Figure 3.1: Hardware architecture with signal flow

ADXL335 sensor breakout, which could potentially make this procedure slower. The highest data volume over a predefined time span achieved for this project is 300Hz.

In essence, a batch of data is received through this procedure by a Raspberry Pi 4, which in turn processes and analyzes these samples.

3.3.2 Scalability Considerations

- **Multi-Threading:** RPi4 handles concurrent sensor inputs (e.g., 4x Arduino nodes).
- **Data Compression:** FFT bin reduction (0–300Hz only) minimizes cloud costs.

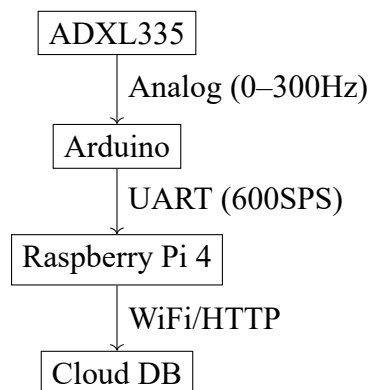


Figure 3.2: End-to-end data flow with protocols

3.3.3 Arduino sketch

Elaboration on (including the whole code):

- Description of sketch and cpp files
- elaborate on how it shares data via UART
- Analyse the possibility of translating Voltage into g or m/s²

3.3.4 Raspberry Pi coding

Elaboration on (including some parts of the code):

- Description of the whole program
- how it posts data via HTTP
- Data stored on database , and database setup

3.3.5 Vercel coding

Elaboration on (including some parts of the code):

- page file that prints data
- how it is coupled with supabase

Chapter 4

Results

4.1 Intro

- depict data dashboard
- analyse alarm cases
- database - stored data

Chapter 5

Conclusions

Εδώ εξηγούμε ότι θα συνοψίσουμε την μελέτη που εκπονήθηκε στα πλαίσια της διπλωματικής.

- summarize how it decides if a bearing is defective
- how it collects data
- how data is analysed and at what stages it is analysed

5.1 Σύνοψη και συμπεράσματα

Εδώ συνοψίζουμε τα αποτελέσματα της διπλωματικής και περιγράφουμε τα συμπεράσματα που προέκυψαν, αρνητικά και θετικά. Επιβεβαιώνουμε την συνεισφορά της διπλωματικής στα προβλήματα που αναφέραμε στην εισαγωγή.

5.2 Μελλοντικές επεκτάσεις

Εδώ δίνουμε ιδέες για επέκταση της διπλωματικής.

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APPENDICES

Appendix

Τίτλος Παραρτήματος

Τα παραρτήματα περιλαμβάνουν συνοδευτικό, υποστηρικτικό υλικό (πίνακες, φωτογραφίες, ερωτηματολόγια, στατιστικά στοιχεία, αποδείξεις, περιγραφές λογισμικών προγραμμάτων, παραδείγματα, περιγραφές πολύπλοκων διαδικασιών, λίστα με πρωτογενή στοιχεία, λεπτομερής περιγραφή και προδιαγραφές εξοπλισμού, οδηγίες εγκατάστασης λογισμικού, κ.λπ.), ή αλλιώς ό,τι θεωρείται χρήσιμο να περιγραφεί, αλλά δεν συνηθίζεται να εντάσσεται μέσα στο κυρίως κείμενο της Εργασίας. Στο κυρίως κείμενο της Εργασίας πρέπει να γίνονται οι κατάλληλες παραπομπές προς τα παραρτήματα, όπου το κείμενο σχετίζεται με υλικό που περιλαμβάνεται σε αυτά. Ένα παράρτημα, αναλόγως με το περιεχόμενό του, μπορεί να είναι ενιαίο, ή να χωρίζεται σε ενότητες.

1 Δυνατότητες του $\text{\LaTeX}$

Καθώς το παρόν αποτελεί ένα πρότυπο συγγραφής διπλωματικών εργασιών, στην ενότητα αυτή επιδεικνύονται ορισμένες από τις δυνατότητες το $\text{\LaTeX}$ οι οποίες μπορούν να αξιοποιηθούν στο κείμενο μιας διπλωματικής εργασίας (μια καλή πηγή για τη διερεύνηση των δυνατοτήτων του $\text{\LaTeX}$ είναι η ιστοσελίδα https://www.overleaf.com/learn/latex/Main_Page).

1.1 Πίνακες

Ο Πίνακας .1 είναι ένα παράδειγμα πίνακα σχεδιασμένου με εντολές του περιβάλλοντος `tabular`.

Table .1: Παράμετροι πειραμάτων

Πλήθος κελιών καννάβου $c \times c$	50×50 , 100×100 , 200×200 , 250×250 , 500×500 , 1000×1000
Τυπική απόκλιση σ	25m, 50m, 75m, 100m , 150m, 200m
Αριθμός εγγύτερων γειτόνων k	1, 2, 3 , 4, 5, 10, 20
Πιθανοτικό κατώφλι θ	50%, 60%, 70%, 75% , 80%, 90%, 99%

1.2 Διαγράμματα - Γραφικές παραστάσεις

Στο Σχήμα .1, παρουσιάζεται ένα παράδειγμα γραφικής παράστασης σχεδιασμένης με το Gnuplot.

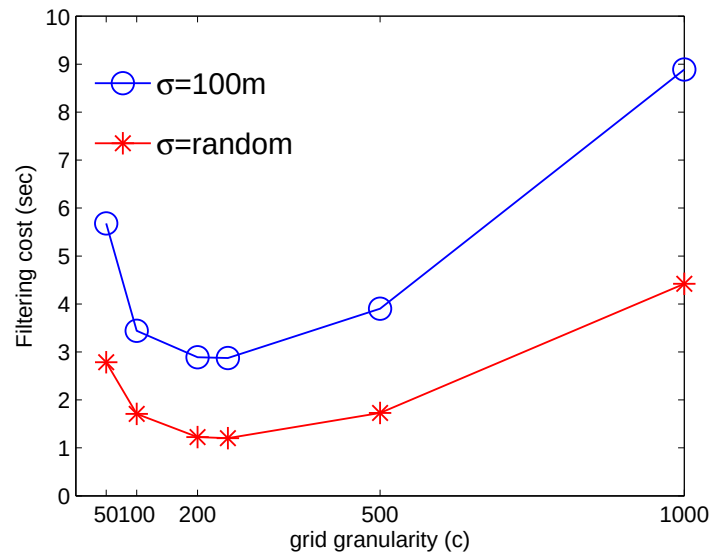


Figure .1: Κλιμάκωση χρόνου εκτέλεσης για διάφορες υποδιαιρέσεις του καννάβου

1.3 Σχήματα

Ακολουθεί στο Σχήμα .2 ένα παράδειγμα σχήματος φτιαγμένου με εντολές του πακέτου TikZ.

1.4 Αλγόριθμοι

Ακολουθεί ο Αλγόριθμος 1, ο οποίος είναι μορφοποιημένος με τα πακέτα algorithm και algorithmic.

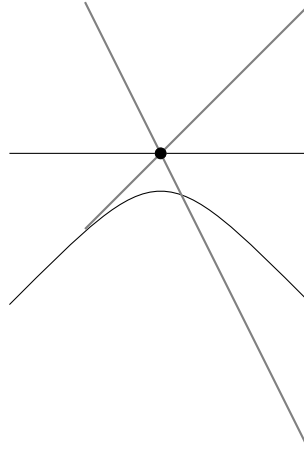


Figure .2: Παράδειγμα σχήματος με εντολές του πακέτου TikZ

Algorithm 1 Probabilistic $k\theta NN$ Monitoring

-
- 1: **Procedure** *VerifyCandidate* (focal query point q , threshold θ , object o , list of auxiliary objects P , distance $kMAXDIST$)
 - 2: **if** $\Phi(o, kMAXDIST) \geq \theta$ **and** $L_2(q, o) \leq L_2(q, P.top())$ **then**
 - 3: $P.pop()$; *//Replace the most extreme element in P , since candidate o ...*
 - 4: $P.push(o)$; *//... has enough probability and has its mean closer to focal q*
 - 5: **end if**
 - 6: **End Procedure**
-

1.5 Μαθηματικές εκφράσεις

Ακολουθούν παραδείγματα μαθηματικών εκφράσεων.

$$\hat{I}(x, u, t) = dist(y(t_f), \Gamma) + \int_t^{t_f} \mathcal{L}(y(s), u(s), s) ds \quad (.1)$$

$$\frac{d}{dx} \left(\int_0^z f(u) du \right) = f(x).$$

1.6 Θεωρήματα, Πορίσματα, Ορισμοί, κλπ.

Ακολουθεί παράδειγμα θεωρήματος από την ιστοσελίδα https://www.overleaf.com/learn/latex/Theorems_and_proofs

Theorem 5.1. *Let f be a function whose derivative exists in every point, then f is a continuous function.*

1.7 Απαριθμήσεις

Μια απαρίθμηση (itemized list) βοηθά στην παρουσίαση μιας σειράς περιπτώσεων με σαφήνεια. Ακολουθεί παράδειγμα.

Η εκπαίδευση στην Ελλάδα διακρίνεται σε:

- Πρωτοβάθμια
- Δευτεροβάθμια
- Τριτοβάθμια

1.8 Είδη πηγών στις αναφορές

Στο references.bib μπορεί να δει κανείς πώς γράφονται διάφορα είδη πηγών (Βιβλία Ξε-νόγλωσσα [13], Βιβλία Ελληνικά [14], Άρθρα σε επιστημονικά περιοδικά [15], Άρθρα σε επιστημονικά συνέδρια [16], Ιστοσελίδες [17], Πτυχιακές Εργασίες [18], Διπλωματικές Εργασίες [19], Μεταπτυχιακές Διπλωματικές Εργασίες [20], Διδακτορικές Διατριβές [21], Τεχνικές Αναφορές [22], Διπλώματα Ευρεσιτεχνίας [23]), Κεφάλαια σε συλλογικούς τόμους [24].